Quiz — Week 9, Class 1

Centralized Multi-Agent Patterns

Name: ______________________________ **Date:** ____________

Week 9 · Class 1 — Centralized Multi-Agent Patterns

> ℹ Short answer — **2–4 sentences each.** Explain your reasoning and how the idea applies to agentic design; definitions alone will not score.

1. Module 9 catalogs the centralized patterns — Orchestrator-Worker, Hierarchical, Supervisor-Router, Sequential Pipeline. Take a task that is a genuinely *fixed* sequence (extract → transform → summarize → cite) and one where the next sub-task *depends unpredictably* on prior results, and say which centralized pattern fits each — then explain why centralized control trades flexibility for predictability and debuggability.

2. Module 9 warns about the **coordination tax** — every added agent multiplies delegation calls, latency, failure modes, and debugging difficulty. You are tempted to add a fourth worker to an orchestrator-worker system. Explain what specific benefit would have to materialize for that fourth agent to be worth the tax, and name a failure mode (e.g., underspecified sub-tasks, lossy aggregation, cascading failure) you would be inviting by adding it.

3. Primers 19 and 20 present MCP and FastMCP as the standardized way agents get tools. In a centralized multi-agent system where several workers each need overlapping capabilities, explain why exposing those capabilities through a *shared MCP tool protocol* (rather than hand-wiring tools into each agent) matters — and what a FastMCP server buys you when you want to swap or version a tool without rewriting every worker.